

Application of Normalized importance sampling in Reward-based sampling from forward sampling models

Suppose we have a distribution $P(\mathbf{x}) = P(x_1, \dots, x_n)$ from which we can easily sample using the forward sampling method.

Now your goal is to impose a cost on $\mathbf{x}$ and sample from

$$P_C(\mathbf{x}) \propto P(\mathbf{x}) \exp(-C(\mathbf{x}))$$

We can use importance sampling as follows:

choose $Q(\mathbf{x}) = P(\mathbf{x})$

$$x \sim \frac{P_C(x)}{P(x)}$$

$$P(x) \quad N \gg M.$$

1) Generate $L \times M = N$ samples from $P(x)$

$x^1, x^2, x^3, \dots, x^N$

$$\frac{P_C(x^1)}{P(x^1)}$$

$$\frac{P_C(x^2)}{P(x^2)}$$

$$\frac{P_C(x^N)}{P(x^N)}$$

$$2) e^{-C(x^1)}$$

$$e^{-C(x^2)}$$

$$\dots e^{-C(x^N)}$$

3) Sample M images from the N examples above
with probability of including i^{th} image
as

$$\frac{e^{-C(x^i)}}{\sum_{j=1}^N e^{-C(x^j)}}$$

Application of Normalized importance sampling in conditional generation from forward sampling models

- 1 Suppose we have a distribution $P(\mathbf{x}) = P(x_1, \dots, x_n)$ from which we can easily sample using the forward sampling method.
- 2 Examples of such distributions: Bayesian networks, diffusion model, VAE, etc.
- 3 Now your goal is to sample from $P(\mathbf{x} | \mathbf{x}_E = \mathbf{v})$ where some subset of values are constrained to take a fixed value. Without loss of generality assume that the condition is only on the last node $x_n = 1$
- 4 How can you modify the forward sampling algorithm to do the above conditional sampling?

$$P(x_1, x_2, \dots, x_{n-1} | x_n = 1)$$

A wrong algorithm for sampling from $P(\mathbf{x} | x_n = 1)$

Do normal forward sampling on $n - 1$ variables, and just replace the variable of x_n to 1.

1 For $i = 1$ to M

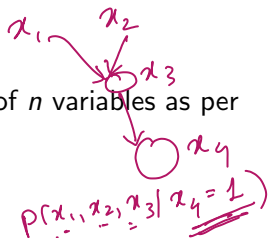
1 Let $j_1, \dots, j_n$ denote a topological sort order of n variables as per G

2 For $k = 1$ to $n - 1$

1 $\xi_{j_k}^i \sim P(x_{j_k} | Pa(j_k) = \xi_{Pa(j_k)}^i)$

$\xi_n^i = 1$

3 Return $\xi^1 \dots \xi^M$



$x_1, x_2, \dots, x_{n-1}$ are samples from the original unconditional distribution $P(x_1, \dots, x_n)$ not from the desired conditional distribution.

Correct fix: Use normalized importance sampling.

We want to sample from

$$\begin{aligned} R(x_1, \dots, x_{n-1}) &= P(x_1, \dots, x_{n-1} | x_n = 1) \\ R(x_1, \dots, x_{n-1}) &= \frac{P(x_1, \dots, x_{n-1}, x_n = 1)}{\sum_{x'_1, \dots, x'_{n-1}} P(x'_1, \dots, x'_{n-1}, x_n = 1)} \end{aligned}$$

Choose the proposal distribution is as

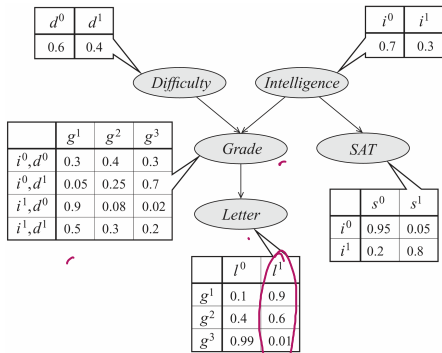
$$Q(x_1, \dots, x_{n-1}) = P(x_1, \dots, x_{n-1})$$

Importance weight then becomes:

$$\frac{R(x_1, \dots, x_{n-1})}{Q(x_1, \dots, x_{n-1})}$$

$$\begin{aligned} &= \frac{P(x_1, x_2, \dots, x_{n-1}, x_n = 1)}{\left(\sum_{x'_1, \dots, x'_{n-1}} P(x'_1, \dots, x'_{n-1}, x_n = 1) \right) P(x_1, x_2, \dots, x_{n-1})} \\ &= \frac{P(x_1, \dots, x_{n-1}) P(x_n = 1 | x_1, x_2, \dots, x_{n-1})}{\sum_{x'_1, \dots, x'_{n-1}} P(x'_1, \dots, x'_{n-1}) P(x_n = 1 | x'_1, \dots, x'_{n-1})} \end{aligned}$$

Example



$$= \frac{P(x_n = 1 | x_1, \dots, x_{n-1})}{Z}$$

Calculate expected SAT score given letter=1

Let $s_0 = 50, s_1 = 100$

$$f(D, I, G, S) = \begin{cases} 50 & \text{if } S = s^0 \\ 100 & \text{if } S = s^1 \end{cases}$$

$$P(D, I, G, S | L=1) \rightarrow$$

$$\sum_{L=1} f(D, I, G, S) P(D, I, G, S | L=1)$$

Example

- * Obtain samples from the proposal distribution : $P(D, I, G, S) \propto P(L=1)$ using forward sampling
- * Weight of a sample
$$P(L=1 | D, I, G, S) = \frac{P(L=1 | G)}{P(L=1)}$$

Limitation of Importance sampling:

if weight $P(x_n=1 | x_{-n})$ is small, most of the samples will not be useful.

Example

Normalized importance sampling is consistent

The normalized importance sampling estimation is:

$$\mu_f(S_Q) = \frac{\frac{1}{M} \sum_i f(\mathbf{x}^i) \tilde{P}(\mathbf{x}^i) / Q(\mathbf{x}^i)}{\frac{1}{M} \sum_i \tilde{P}(\mathbf{x}^i) / Q(\mathbf{x}^i)}$$

$$\begin{aligned} & \rightarrow \frac{\sum_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \tilde{P}(\mathbf{x})}{Z} \\ & Z = \sum_{\mathbf{x} \in \mathcal{X}} \tilde{P}(\mathbf{x}) \end{aligned}$$

As $M \rightarrow \infty$, the RHS becomes

$$\begin{aligned} & \frac{\sum_{\mathbf{x} \in \mathcal{X}} Q(\mathbf{x}) [f(\mathbf{x}) \tilde{P}(\mathbf{x}) / Q(\mathbf{x})]}{\sum_{\mathbf{x} \in \mathcal{X}} Q(\mathbf{x}) [\tilde{P}(\mathbf{x}) / Q(\mathbf{x})]} = \frac{\sum_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \tilde{P}(\mathbf{x})}{\sum_{\mathbf{x} \in \mathcal{X}} \tilde{P}(\mathbf{x})} \\ & = \sum_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \frac{\tilde{P}(\mathbf{x})}{Z} = E_P[f(\mathbf{x})] \end{aligned}$$

$$\mu_f(S_Q : |S_Q| = 1) = f(\mathbf{x}^i) \quad \mathbf{x}^i \sim Q(\mathbf{x})$$

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N f(\mathbf{x}^i) =$$

Normalized importance sampling is biased

Assume just one sample per set $M = 1$. Consider a sample $S_i = \{\mathbf{x}^i\}$ of size 1. The estimated mean with this sample is

$$\mu_f(S_i) = \frac{f(\mathbf{x}^i)\tilde{P}(\mathbf{x}^i)/Q(\mathbf{x}^i)}{\tilde{P}(\mathbf{x}^i)/Q(\mathbf{x}^i)} = f(\mathbf{x}^i)$$

When we sample a large number of sets $S_1, \dots, S_N$ each of size 1, then the average of this estimate becomes:

$$\frac{1}{N} \sum_{j=1}^N \mu_f(S_j) = \frac{1}{N} \sum_j f(\mathbf{x}^j)$$

Since the elements in each S_j are sampled from the $Q(\mathbf{x})$ distribution, the above as $N \rightarrow \infty$ will converge to

$$\sum_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})Q(\mathbf{x}) = E_Q[f] \neq E_P[f] \text{ unless } P=Q$$

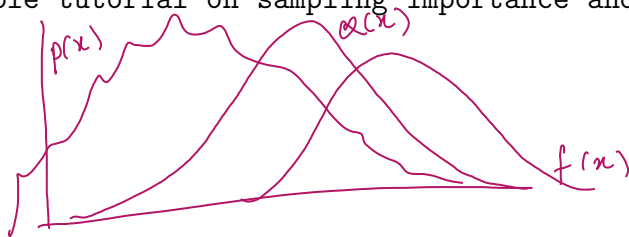
Importance sampling: Choosing $Q(x)$

The choice of $Q(x)$ for which the expected error of the estimate is minimum is when

$$\underline{Q(x)} \propto \underline{|f(x)|P(x)}, \quad Q(x) > 0 \text{ whenever } P(x) > 0. \quad (2)$$

Normalized importance sampling is biased when M is small.
Designing a good $Q(x)$ for which the sampling is efficient is not always easy.

A gentle tutorial: https://medium.com/@amir_masoud/a-simple-tutorial-on-sampling-importance-and-monte-carlo-



MCMC: Markov Chain Monte Carlo Sampling

When designing proposal distribution is not easy.

Goal: sample from $P(\mathbf{x} = x_1, \dots, x_n)$ Easy to sample either

- 1 Conditional probability of one variable: $P(x_i | \mathbf{x}_{-i})$ (Gibbs samplings)
- 2 Ratio of two probabilities: $P(\mathbf{x})/P(\mathbf{x}')$ — normalizer not required. (Metropolis Hastings)
- 3 Gradient of $\log P(\mathbf{x})$ wrt $\mathbf{x}$. (Langevin monte carlo)

Useful when all else fails, guaranteed to converge to the optimal over infinite number of samples.

MCMC Sampling Transition function

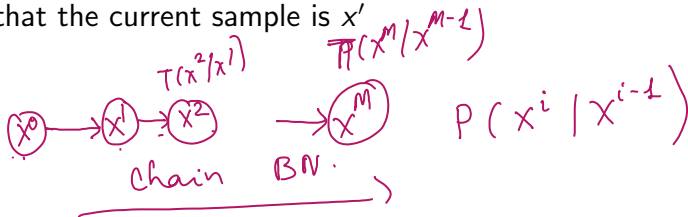
Designed by us much like the proposal distribution in Importance Sampling

$$\sum_{x \in \mathcal{X}} \underline{T(x|x')} = \underline{1}$$

where

$$\underline{T(x|x')} \geq 0 \quad \forall \quad x, x' \in \mathcal{X}$$

$T(x|x')$ denotes the probability that the next sampled value is x given that the current sample is x'

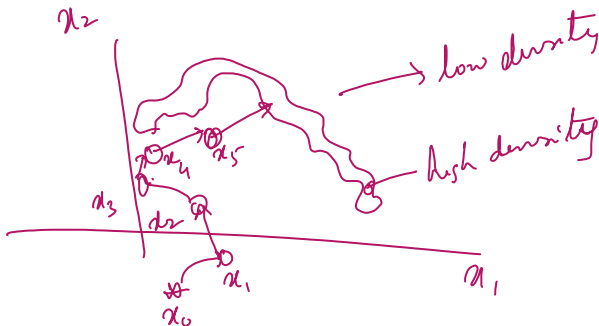


MCMC Sampling Algorithm

- 1 Start with an initial sample x^0
- 2 For $t = 1$ to a large number L
 $x^{t+1} \sim T(x|x' = x^t)$
 $x^0 \rightarrow x^1 \rightarrow x^2 \rightarrow \dots \rightarrow x^L$
- 3 Actually perform the sampling for $t = L+1$ to $t = L+M$
 $x^t \sim T(x|x^{t-1})$
- 4 Return $x^{L+1}, x^{L+2}, \dots, x^{L+M}$ as samples

Intuition behind MCMC

- 1 The $P(\mathbf{x})$ distribution is complicated but the regions of high probability form a locally connected space (manifold).
- 2 We design T so that we stay in the manifold and starting from any random point in $\mathcal{X}$ we are able to reach the manifold.



Gibbs sampling demo

[https://chi-feng.github.io/mcmc-demo/app.html?
algorithm=GibbsSampling&target=banana](https://chi-feng.github.io/mcmc-demo/app.html?algorithm=GibbsSampling&target=banana)

Example

$p(x)$

$n = 1, \mathcal{X} = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$

Pictorial depiction of $T(x|x')$



$P_t(X^t = v)$ probability that after t steps of MCMC sampling using the above $T(x|x')$ the sampled value will be v .

First sample is $X^0 = 0$ $P(X^0 = 0) = 1$

$t=1$ $P(X^1 = x)$ $P(X^1 = -1) = 0.25$; $P(X^1 = +1) = 0.25$
 $P(X^1 = 0) = 0.5$

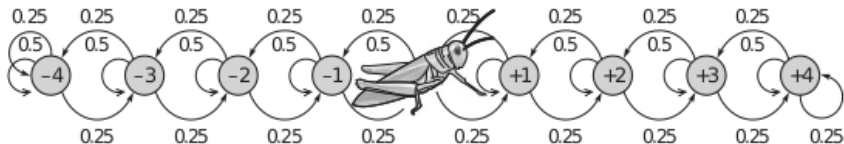
$$t=2: P(X^2 = x) = \sum_{x'} P(X^2 = x | X^1 = x') P(X^1 = x')$$

$$P(X^2 = -2) = 0.25 \times 0.25$$

$$P(X^2 = 0) = P(X^2 = 0 | X^1 = 0) P(X^1 = 0) + P(X^2 = 0 | X^1 = +1) P(X^1 = +1) + P(X^2 = 0 | X^1 = -1) P(X^1 = -1)$$

Example

$$= \frac{3}{8}$$



$$t=0 \quad P_{t=0}(X^0 = \mathbf{x}^0) = 1$$

$$P_{t=0}(X^0 = 0) = 1$$

$$t=1 \quad P_{t=1}(X^1 = v) = T(x = v | X' = \mathbf{x}^0)$$

$$P(X^1 = -1) = 0.25, P(X^1 = 0) = 0.5, P(X^1 = 1) = 0.25$$

$$t=2 \quad P_{t=2}(X^2 = v) = \sum_{x'} P_{t=1}(X') T(x = v | x')$$

Continued

$t + 1$ $P_{t+1}(x^{t+1} = v) = \sum_{x'} P_t(x') T(x = v | x')$
 $t \rightarrow \infty$ We want $P_{t+1}(x^{t+1})$ to converge to $P(\mathbf{x}) = P(x_1, x_2, \dots, x_n)$

$P_{t+1}(x) \approx P_t(x)$ for large ts

At convergence as $t \rightarrow \infty$

$P_{t+1}(x) = P_t(x) = \pi(x) \leftarrow$ Stationary Distribution